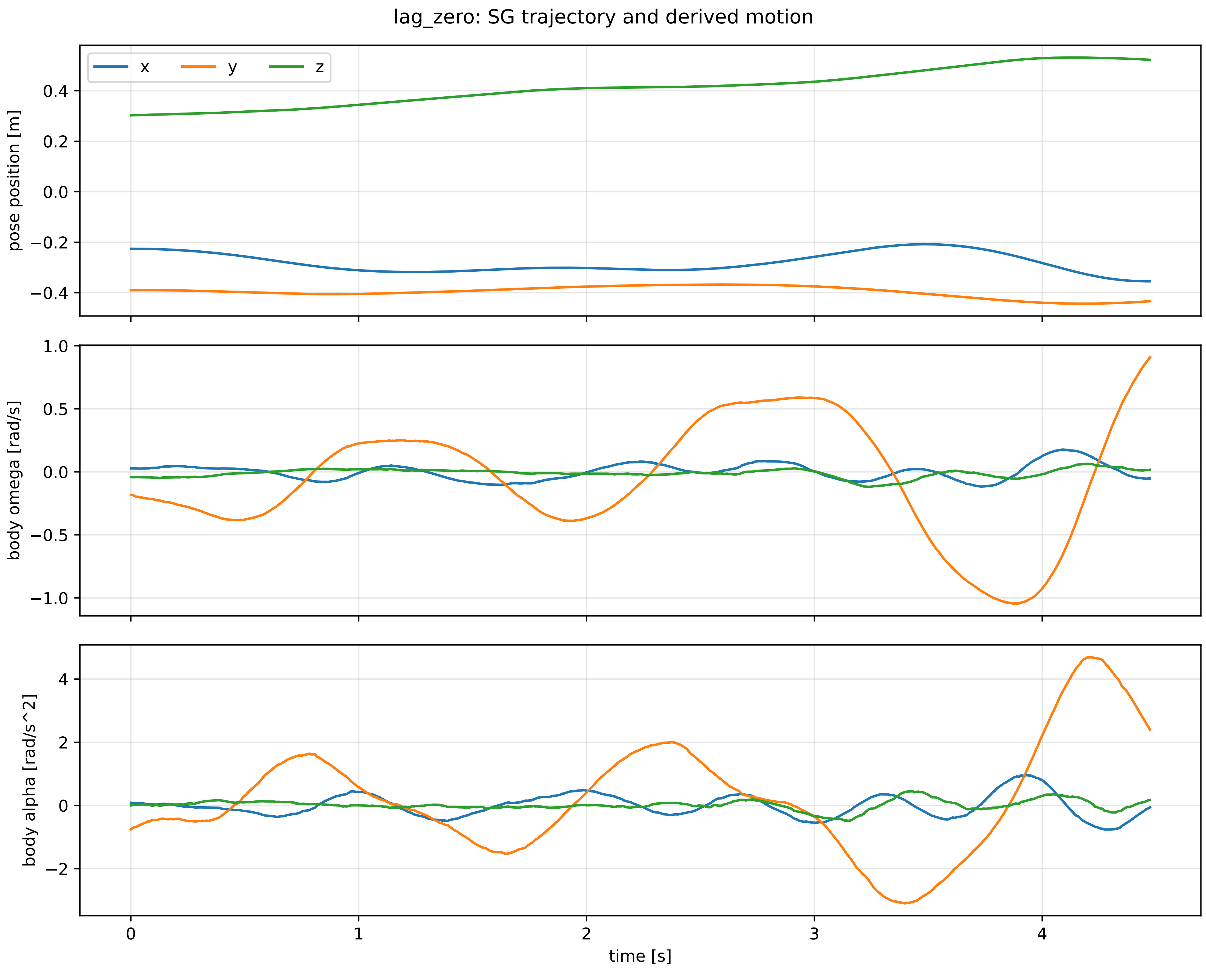
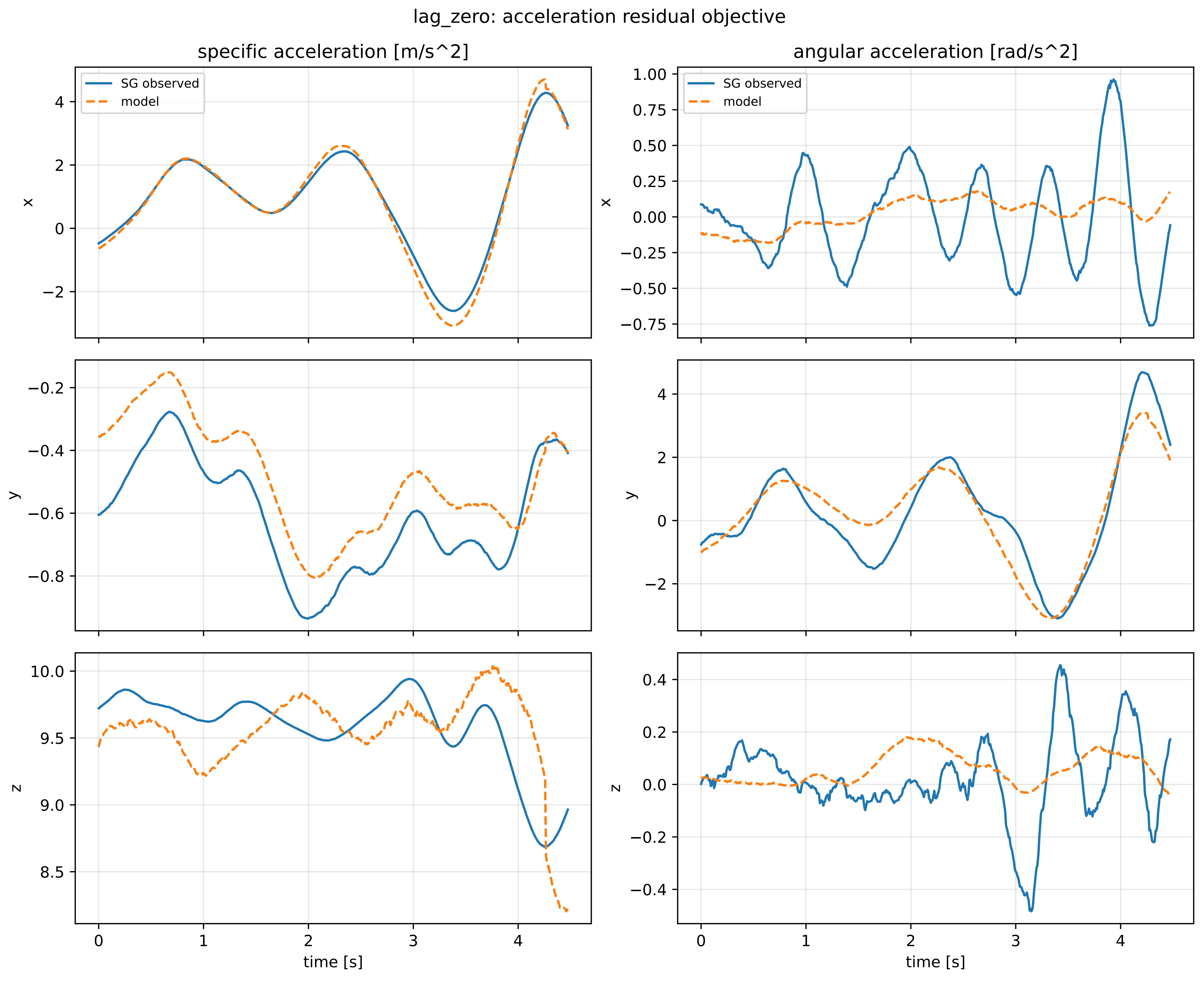
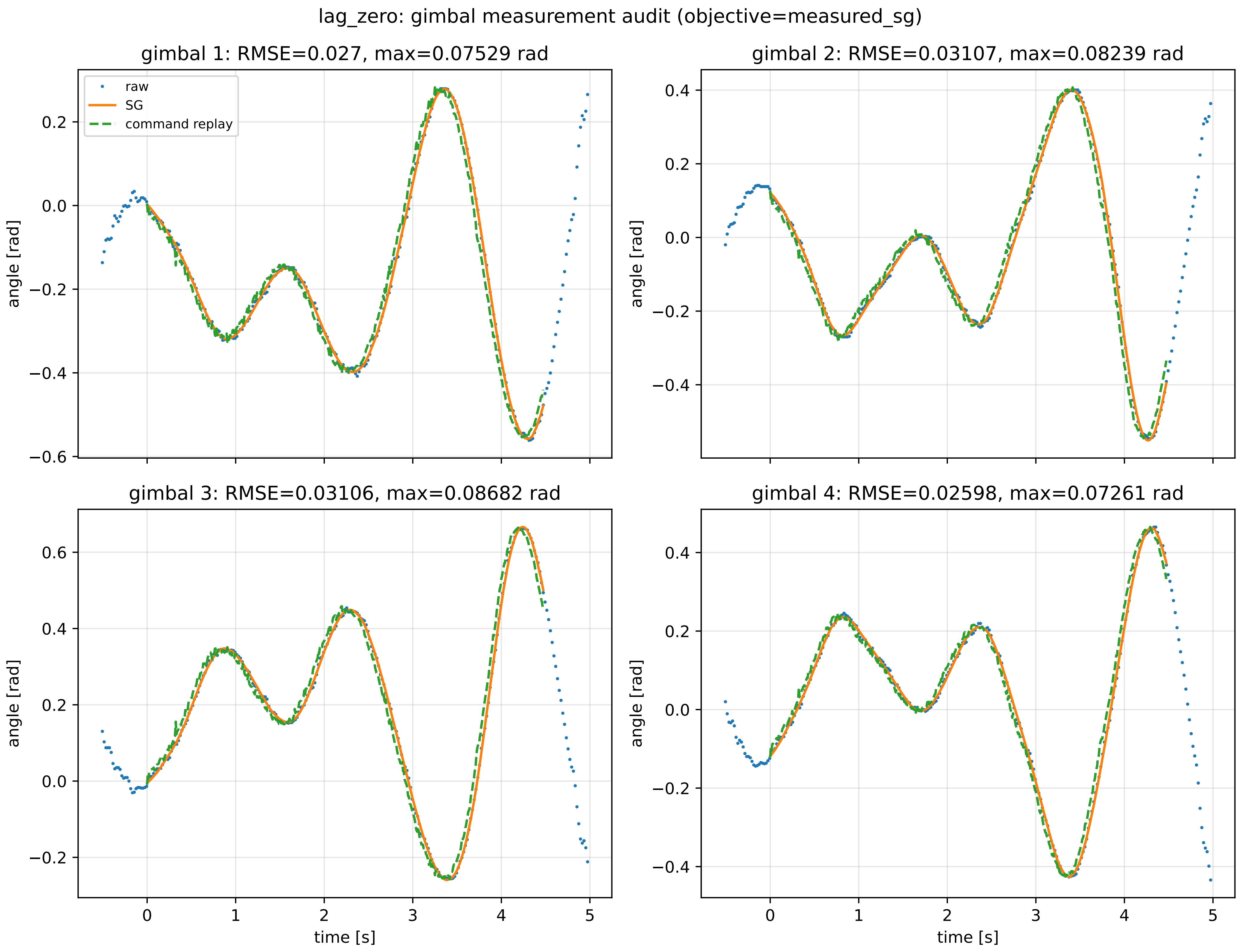
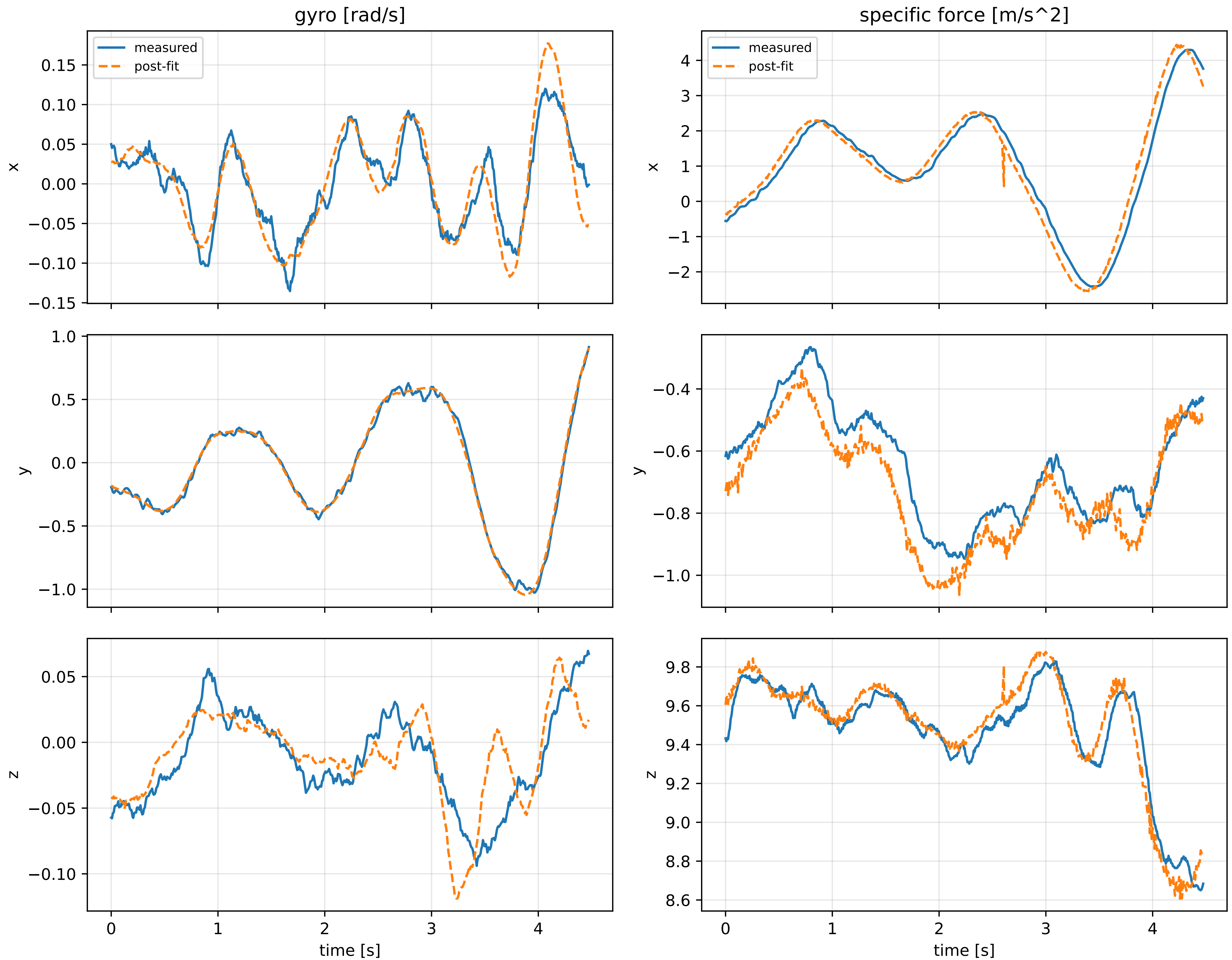


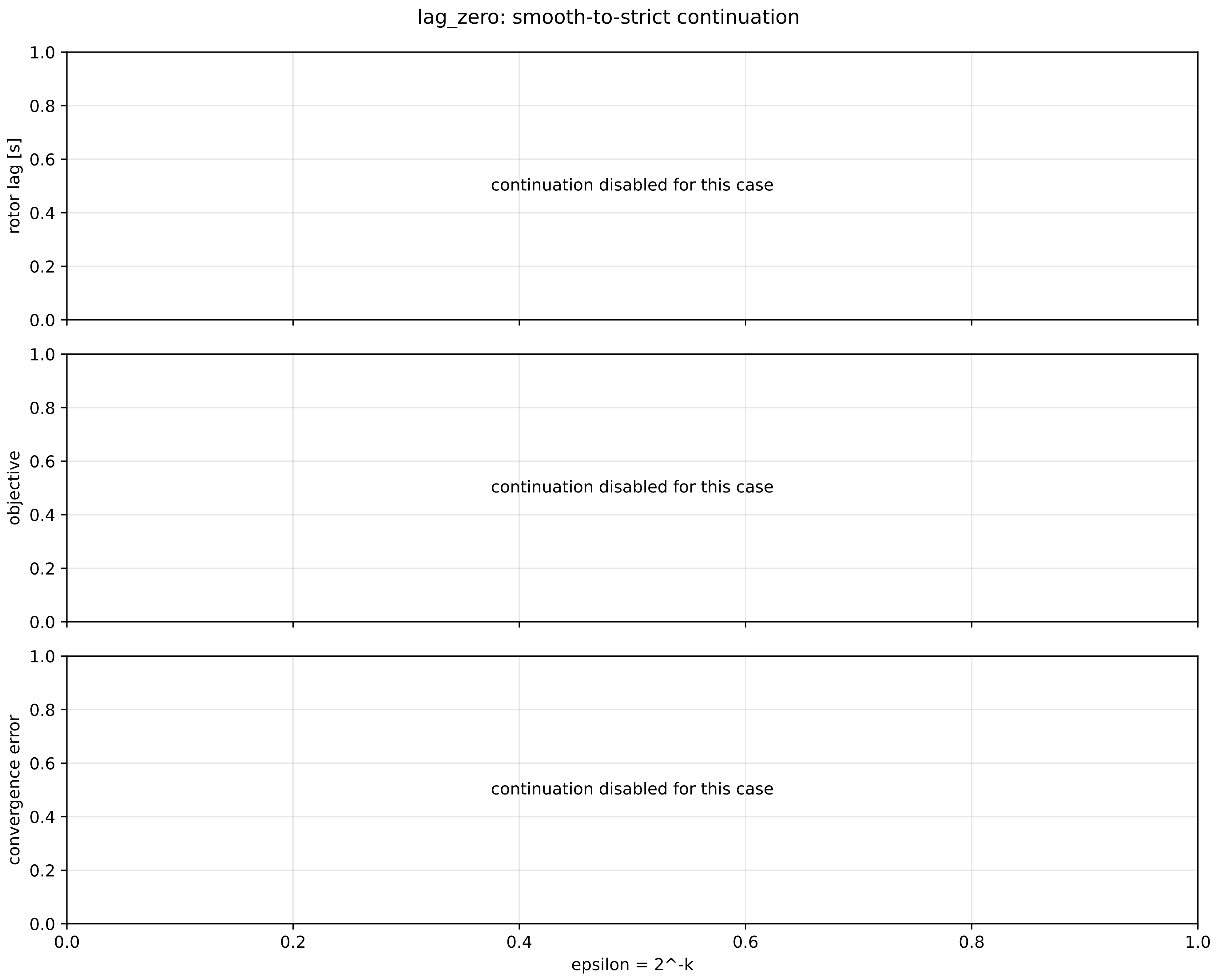
lag_zero: SG trajectory and derived motion
x
y
z
pose position [m]
0.4
0.2
0.0
−0.2
−0.4
body omega [rad/s]
1.0
0.5
0.0
−0.5
−1.0
body alpha [rad/s^2]
4
2
0
−2
0
1
2
3
4
time [s]

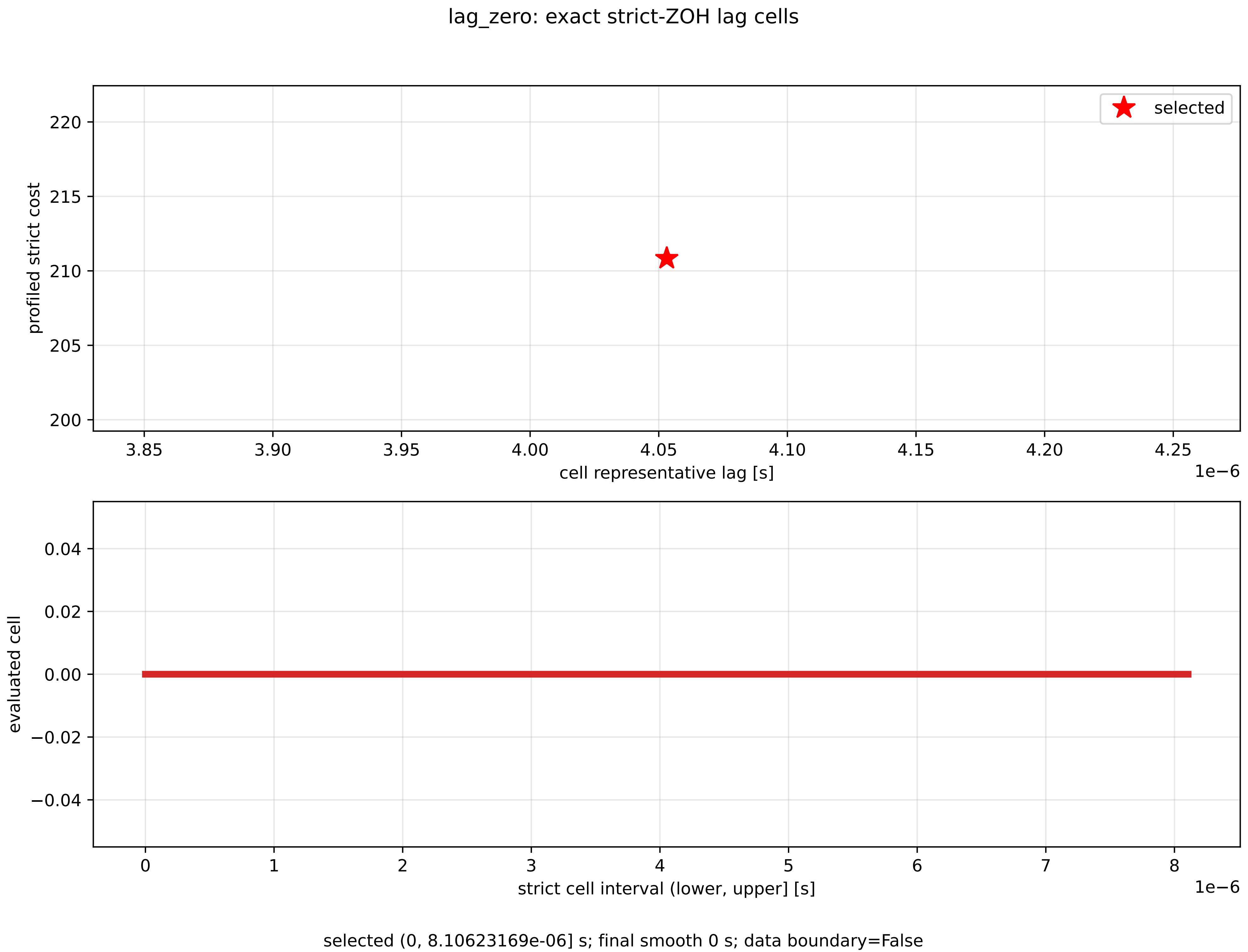

lag_zero: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]


lag_zero: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.027, max=0.07529 rad
gimbal 2: RMSE=0.03107, max=0.08239 rad
gimbal 3: RMSE=0.03106, max=0.08682 rad
gimbal 4: RMSE=0.02598, max=0.07261 rad
raw
SG
command replay
angle [rad]
time [s]

lag_zero: IMU comparison (diagnostic only)

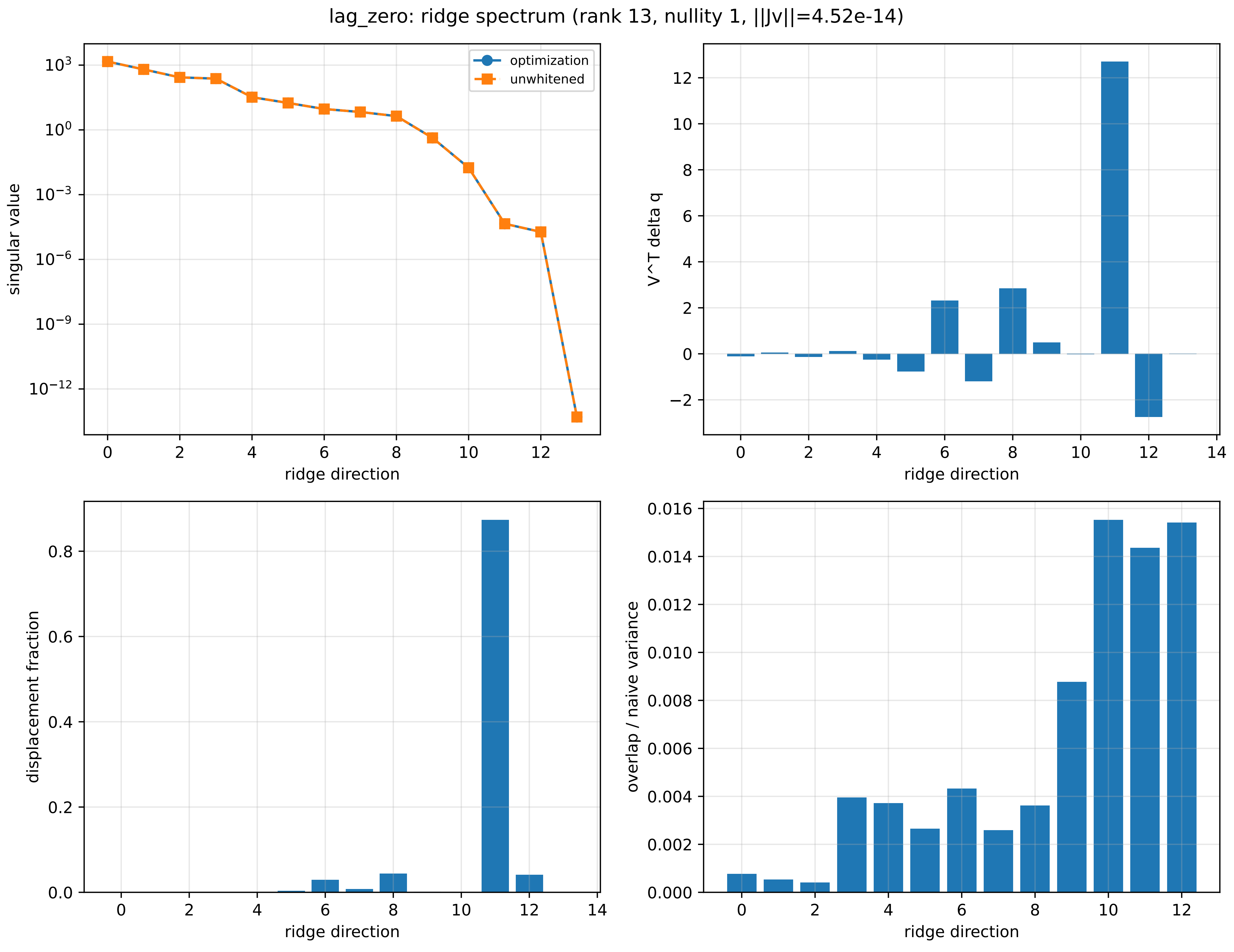


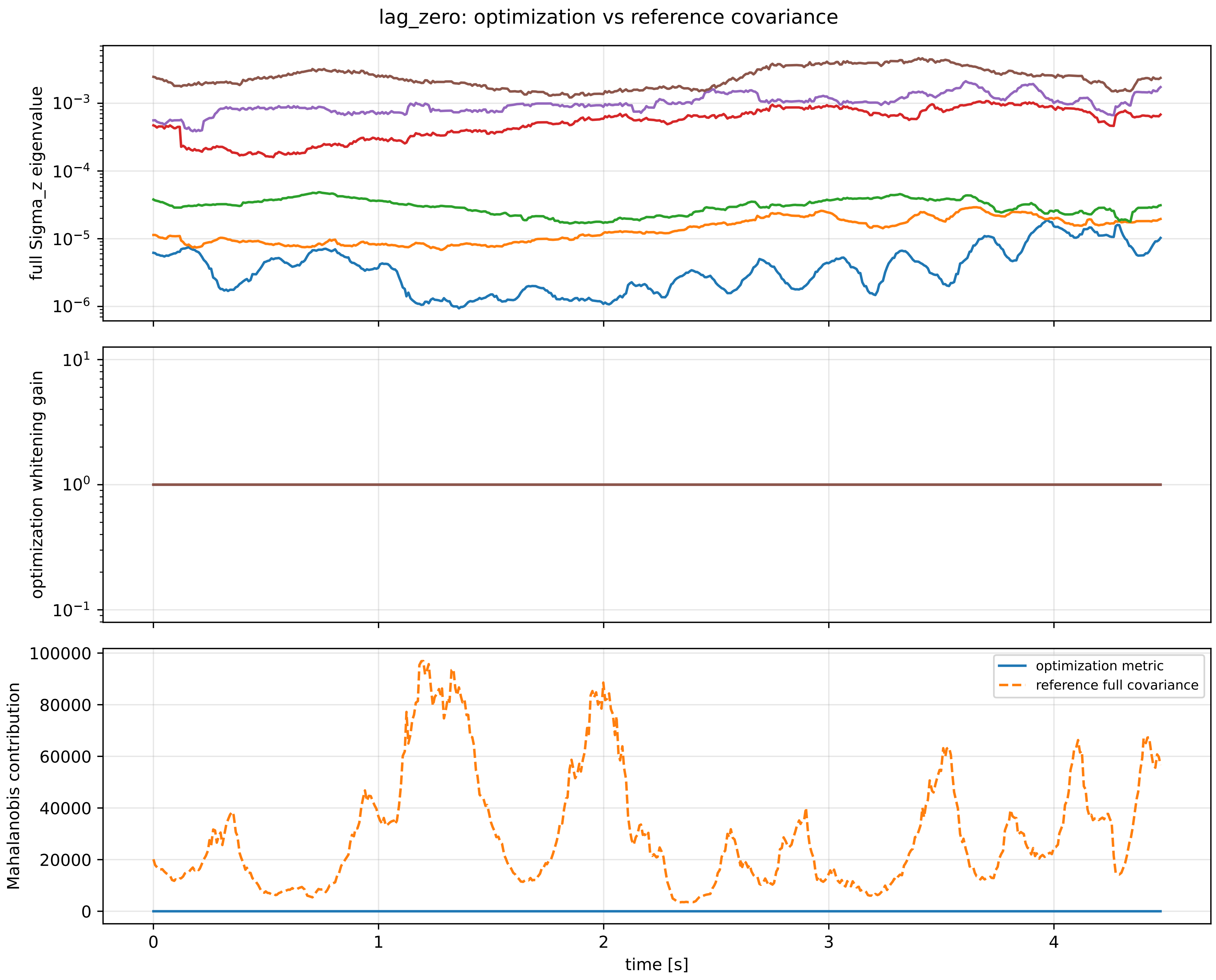

lag_zero: smooth-to-strict continuation
rotor lag [s]
continuation disabled for this case
objective
continuation disabled for this case
convergence error
continuation disabled for this case
epsilon = 2^-k
0.0
0.2
0.4
0.6
0.8
1.0


lag_zero: exact strict-ZOH lag cells
selected
profiled strict cost
220
215
210
205
200
3.85
3.90
3.95
4.00
4.05
4.10
4.15
4.20
4.25
1e−6
cell representative lag [s]
evaluated cell
0.04
0.02
0.00
−0.02
−0.04
0
1
2
3
4
5
6
7
8
1e−6
strict cell interval (lower, upper] [s]
selected (0, 8.10623169e-06] s; final smooth 0 s; data boundary=False

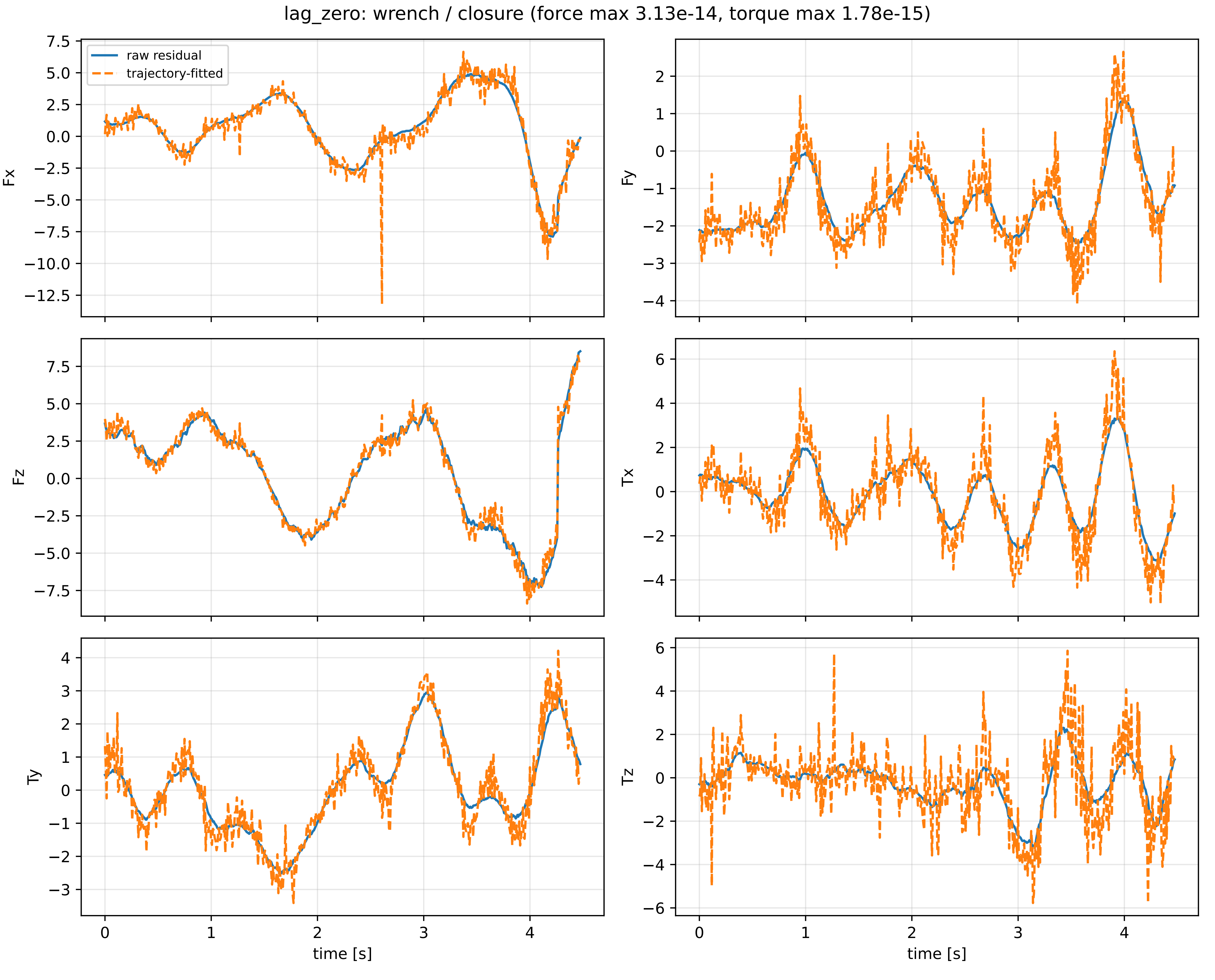
lag_zero: parameter estimate

```
Case: lag_zero
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 10.6294881
inertia [kg m^2]:
[[ 3.92355568 -0.13694608 -0.05490876]
 [-0.13694608  1.81761612 -0.66729721]
 [-0.05490876 -0.66729721  5.50598531]]
CoG body [m]: [-0.17085214  0.0779173  -0.17037739]
force effectiveness: [5.19182203e+00 2.40212073e-05 2.19830575e+00 7.52740611e+00]

scale-free J/m [m^2]:
[[ 0.36911991 -0.0128836  -0.0051657 ]
 [-0.0128836   0.17099752 -0.06277793]
 [-0.0051657  -0.06277793  0.51799158]]
scale-free f/m: [4.88435755e-01 2.25986493e-06 2.06812006e-01 7.08162620e-01]

rotor lag cell: (0, 8.10623169e-06] s
representative: 0 s
```


lag_zero: ridge spectrum (rank 13, nullity 1, ||Jv||=4.52e-14)
optimization
unwhitened
singular value
ridge direction
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction



lag_zero: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-3}
10^{-4}
10^{-5}
10^{-6}
optimization whitening gain
10^{1}
10^{0}
10^{-1}
Mahalanobis contribution
100000
80000
60000
40000
20000
0
optimization metric
reference full covariance
0
1
2
3
4
time [s]


lag_zero: wrench / closure (force max 3.13e-14, torque max 1.78e-15)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]